

Problem 14.2.9

Integrate $f(z)$ counterclockwise around the unit circle. Indicate whether Cauchy's integral theorem applies. Show the details.

$$f(z) = \exp(-z^2)$$

Solution

f is entire since f' is defined everywhere, therefore by Cauchy's integral theorem

$$\oint_C f(z) dz = 0$$

Problem 14.2.11

Integrate $f(z)$ counterclockwise around the unit circle. Indicate whether Cauchy's integral theorem applies. Show the details.

$$f(z) = \frac{1}{2z - 1}$$

Solution

f' is not defined at $z = \frac{1}{2}$ and therefore Cauchy's integral theorem does not apply.

$$C : z = e^{it}, dz = ie^{it} dt, \quad t \in [0, 2\pi]$$

$$\begin{aligned} \oint_C f(z) dz &= \oint_C f(z(t)) \dot{z}(t) dt \\ &= \int_0^{2\pi} \frac{ie^{it}}{2e^{it} - 1} dt \end{aligned}$$

$u = 2e^{it} - 1$, $du = 2ie^{it} dt$, C' is the circle centered at -1 with radius 2

$$\begin{aligned} \int_0^{2\pi} \frac{ie^{it}}{2e^{it} - 1} dt &= \frac{1}{2} \oint_{C'} \frac{1}{u} du \\ &= \frac{1}{2} \ln u \Big|_{C'} \\ &= \frac{1}{2} [\ln |u| + i \arg(u)]_{t=0}^{t=2\pi} \\ &= \boxed{\pi i} \end{aligned}$$

Problem 14.2.13

Integrate $f(z)$ counterclockwise around the unit circle. Indicate whether Cauchy's integral theorem applies. Show the details.

$$f(z) = \frac{1}{z^4 - 1.1}$$

Solution

f' is not defined at $z = \sqrt[4]{1.1} \approx 1.02$, however this is outside the unit circle and therefore Cauchy's integral theorem applies.

$$\oint_C f(z) dz = 0$$

Problem 14.2.18

Integrate $f(z)$ counterclockwise around the unit circle. Indicate whether Cauchy's integral theorem applies. Show the details.

$$f(z) = \frac{1}{4z - 3}$$

Solution

f' is not defined at $z = \frac{3}{4}$ and therefore Cauchy's integral theorem does not apply.

$$C : z = e^{it}, dz = ie^{it} dt, \quad t \in [0, 2\pi]$$

$$\begin{aligned} \oint_C f(z) dz &= \oint_C f(z(t)) \dot{z}(t) dt \\ &= \int_0^{2\pi} \frac{ie^{it}}{4e^{it} - 3} dt \end{aligned}$$

$u = 4e^{it} - 3$, $du = 4ie^{it} dt$, C' is the circle centered at -3 with radius 4

$$\begin{aligned} \int_0^{2\pi} \frac{ie^{it}}{4e^{it} - 3} dt &= \frac{1}{4} \oint_{C'} \frac{1}{u} du \\ &= \frac{1}{4} \ln u \Big|_{C'} \\ &= \frac{1}{4} [\ln |u| + i \arg(u)]_{t=0}^{t=2\pi} \\ &= \boxed{\frac{\pi i}{2}} \end{aligned}$$

Problem 14.2.22

Evaluate the integral. Does Cauchy's theorem apply? Show details.

$$\oint_C \operatorname{Re} z dz$$

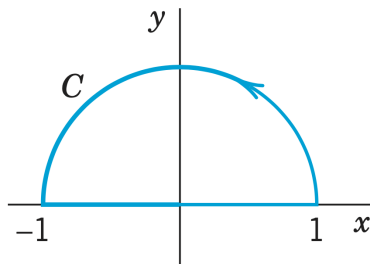


Figure 1: C

Solution

$$f(z) = \operatorname{Re} z = u(x, y) + iv(x, y)$$

$$\begin{aligned} u &= x & v &= 0 \\ u_x &= 1 & v_x &= 0 \\ u_y &= 0 & v_y &= 0 \end{aligned}$$

$u_x \neq v_y$ therefore f is not analytic and Cauchy's theorem does not apply.

$C_1 : z = e^{it}, dz = ie^{it} dt, \operatorname{Re} z = \cos t, \quad t \in [0, \pi]$ $C_2 : z = t, dz = dt, \operatorname{Re} z = t, \quad t \in [-1, 1]$

$$\begin{aligned} \oint_C f(z) dz &= \oint_C f(z(t)) \dot{z}(t) dt \\ &= \int_0^\pi \cos t i e^{it} dt + \int_{-1}^1 t dt \\ &= \int_0^\pi \cos t i (\cos t + i \sin t) dt + \int_{-1}^1 t dt \\ &= i \int_0^\pi \cos^2 t dt - \int_0^\pi \cos t \sin t dt + \int_{-1}^1 t dt \\ &= \frac{i}{2} \int_0^\pi 1 + \cos 2t dt - \frac{1}{2} \int_0^\pi \sin 2t dt + \int_{-1}^1 t dt \\ &= \frac{i}{2} \left[t + \frac{\sin 2t}{2} \right]_0^\pi + \frac{1}{4} [\cos 2t]_0^\pi + \frac{1}{2} [t^2]_{-1}^1 \\ &= \frac{i}{2} [\pi] + \frac{1}{4} [0] + \frac{1}{2} [0] \\ &= \boxed{\frac{\pi i}{2}} \end{aligned}$$

Problem 14.2.24

Evaluate the integral. Does Cauchy's theorem apply? Show details. (*Use partial fraction*)

$$\oint_C \operatorname{Re} z dz$$

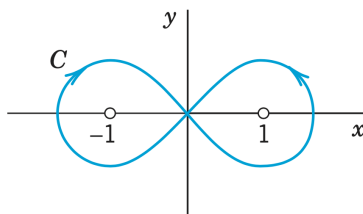


Figure 2: C

Solution

Problem 14.3.3

Integrate by Cauchy's formula counterclockwise around the circle.

$$\frac{z^2}{z^2 - 1}, \quad |z + i| = 1.4$$

Solution

Problem 14.3.11

Integrate counterclockwise.

$$\oint_C \frac{dz}{z^2 + 4}, \quad C : 4x^2 + (y - 2)^2 = 4$$

Solution

Problem 14.3.13

Integrate counterclockwise.

$$\oint_C \frac{z + 2}{z - 2} dz, \quad C : |z - 1| = 2$$

Solution

Problem 14.3.15

Integrate counterclockwise. (compute it for two squares: one with vertices $\pm 2, \pm 2i$; the other one with vertices $\pm 4, \pm 4i$)

$$\oint_C \frac{\cosh(z^2 - \pi i)}{z - \pi i} dz, \quad C \text{ the boundary of the square with vertices } \pm 2, \pm 2, \pm 4i$$

Solution

Problem 14.4.1

Integrate counterclockwise around the unit circle.

$$\oint_C \frac{\sin z}{z^4} dz$$

Solution

Problem 14.4.3

Integrate counterclockwise around the unit circle.

$$\oint_C \frac{e^z}{z^n} dz, \quad n = 1, 2, \dots$$

Solution

Problem 14.4.6

Integrate counterclockwise around the unit circle.

$$\oint_C \frac{dz}{(z - 2i)^2(z - i/2)^2}$$

Solution

Problem 14.4.11

Integrate. Show the details. *Hint.* Begin by sketching the contour. Why?

$$\oint_C \frac{(1+z)\sin z}{(2z-1)^2} dz, \quad C : |z-i| = 2 \text{ counterclockwise}$$

Solution